Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/23/2007
PI Name:	Pearlman, Howard G. Ph.D.		
Project Title:	The Cool Flames Space-Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Howard Pearlman was formerly in the database as a PI at Drexel University. This is the same Howard Pearlman. He is now at GRC working with John McQuillen in PS.		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	09/01/2003	**End Date:**	12/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: per N. Hall (9/06), grant end date should be 12/31/2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1006		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:

Solid fuel and premixed gas combustion exhibits some amazing self-organizaing patterns that rival those previously observed in the famous Belousov-Zhabotinskii liquid phase reaction and biological systems such as the electrical wave propagation observed in potential traces around the human heart. Such self-organizing patterns, including rotating spiral waves and target patterns, can produce instabilities and lead to nonuniform combustion and irregular pressure hstories. This can affect rocket motor thrust vs time profiles. Moreover, this study as much further reaching implications in that it may lead to an improved understanding of self-organizing structures that occur in a variety of chemical and physcal systems. Some have even suggested that such a mechanism may be responsible for LIFE on Earth and perhaps other planets! Understanding self-organizing structures is therefore imperative and will directly support our President's Exploration agenda.
In addition to understanding the mechanisms that lead to self-organization, cool flames are explored since they offer a new way to validate chemical kinetic mechanisms and an improve understanding is needed before newly developed engines (such as Homogeneous Charge Compression Ignition Engines) developed by most auto manufacturers become a reality. These engines offer the high efficiency of a diesel engine with the low emmissions common to spark-ignition engines. Imrpved control, especially at high loads, is important and needs to be addressed before such revolutionary concepts become a reality. To understand cool flames, the Cool Flames Experiment haad successfully passed its science concept review several years back and was scheduled to be developed for the ISS. In the past year, little progress has been made on the development side, while significant progress has been made in modeling these phenomena.

The understanding of self-organization and cool flames are complimentary efforts with different objectives.

The first effort is directed at developing a greater understanding of the high-Lewis number diffusive-thermal instability in ground-based studies, exploring the role of Lewis number, gas mixtures, heat loss and dditives/inhibitors on system stability and reactive-diffusive structure. Experimentally, a large diameter tube filled with a mixture of known gases diluted with helium (one of the most prevalent the most prevalent elements belived to exist in the primordial gas that comprised Earth's atmosphere. The large diameter tube is important because the instability manifests itself as a long wavelength wave on the surface of an otherwise freely-propagating flame as the flame traverses the tube. To eliminate the complexities associated with the buoyant transport processes, reduced gravity testing is suggested to obtain an unobscured view of the competing diffusive processes of mass and heat.

The second effort aims to understand low-temperature oxidation reactions and cool flames. This is a flight-definition program. It considers a static reactor with prescribed mixture of known, premixed gases. The chamber and thus the gases are heated until a reaction, often a precursor to ignition, is initiated spontaneously, without spark or external ignition. In these systems, buoyancy alters the hydrodynamic, thermal and concentration fields. In turn, this alters the flame shape and reaction rate. Since the time required for reaction initiation can be on the order of minutes, or hours or even days(!), these experiments must be performed in long duration reduced gravity facilities. Such reactions have been the cause of extensive fire damage in art galleries, corn silos, grain piles, etc on Earth and it is critical to imporve our understanding of what conditions are safe and what conditions are not safe for short and long duration space flight. The amazing thing with cool flames is that the reactions can be extremely slow and take hours or days before the first sign of a fire is observed! During this time, however, ignition precursors are being formed, which can be detected in advance of the fire (autoignition event), thus preventing a fire in the first place. Prevention via early detection is no less important when it comes to fire safety then it is in medicine or other fields! Early detection saves lives!

Instabilities similar to the diffusive-thermal instability to be studied have been observed in combustion synthesis of materials, as well as a host of other chemical and biological systems, including electrical propagation of signals in and around heart tissue (note, breakup of spatially organized waves is believed to be responsible for heart problems, most notably sudden cardiac death). In material synthesis, burning of a solid is involved, which mandates that the combustion can only be understood on the surface of the material. Since the current investigation involves gaseous components, transport throughout the system can be observed and parameters more easily varied. The knowledge thus gained, which is not available in any other configuration, should be directly applicable to combustion synthesis of materials as well as a providing a fundamental understanding of the couplings between reaction and diffusion, common to all combustion systems. The cool flame investigation is aimed at understanding processes that occur leading to spontaneous ignition. As such, the investigation will lead to an increased understanding of the physics behind knock in engines, which fuel manufacturers avoid by using additives, control of newly developed efficient engines and safety related concerns in spacecraft. This is lacking in current combustion theory.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Refined chemical kinetic models can be baselined using the data obtained from this experimental program. These refined models will be used to more accurately predict the combustion behavior of hydrocarbon fuels at low temperatures. It is relevant to autoignition in newly developed Homogeneous Charge Compression Ignition engines and also useful to mitigate potential fire hazards in space-based platforms. Prevention via early detection is no less important when it comes to fire safety then it is in medicine! Early detection saves lives!
In addition, the Cool Flames Experiment presented an altogether new way for chemists and combustion scientists to validate detailed chemical kinetic models that typically have 100's of parameters (activation energies, pre-exponential factors, temperature exponents) that must be validated and tuned to experimental data. These models can then be used to predict the temperature profiles in engines and the efficiencies that can be achieved as well as the pollutants produced. Unfortunately, NASA terminated this grant prematurely and without access to microgravity facilities, experimental data needed to validate these models cannot be obtained.

Task Progress:	A tremendous amount of model work was conducted and experimental data obtained in KC-135 A flights at microgravity and reduced gravity were analyzed and compared to predictions. Four technical journal papers were written (two appearred in print and two are under review) based on the work conducted. Refer to the Bibliography section below for additional details. The topics covered during this period include: 1. the observation of multiple cool flames at mg and the development of a model that includes diffusive transport of heat and species as well as natural convection that proved able to capture the qualitative behavior; the results suggest that enhanced conduction heat transfer can provide the required heat loss to sustain cool flame oscillations; 2. the effects of gas compressibility on cool flames has been studied numerically; 3. the first quantifiable data on flame propagation speeds of cool flames under pure diffusive conditions has been developed; 4. the effects of diffusive fluxes of heat and species has been studied numerically and experimentally. The reader is referred to the technical papers noted below for additional details. In addition, Fire Science Publishers, a division of Fire Science, Inc., published a text "Ignition Handbook" (ISBN number is: 0-9728111-3-3), authored by Dr. Vytenis (Vyto) Babrauskas. Our cool flame and multi-stage ignition research results obtained at normal and microgravity are highlighted in the text.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Pearlman H. "Multiple cool flames in static, unstirred reactors under reduced-gravity and terrestrial conditions." Combustion and Flame 2007 Mar;148(4):280-4. http://dx.doi.org/10.1016/j.combustflame.2006.11.005 , Mar-2007
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Articles in Peer-reviewed Journals	Fairlie R (School of Chemistry, University of Leeds), Griffiths JF, Hughes KJ, Pearlman H. "Cool flames in space: experimental and numerical studies of propane combustion." Proceedings of the Combustion Institute (Proceedings of the Combustion Institute -Thirtieth International Symposium on Combustion) 2005 Jan;30(1):1057-64. http://dx.doi.org/10.1016/j.proci.2004.08.081 , Jan-2005
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Papers from Meeting Proceedings	Foster M, Pearlman H. "Diffusion-controlled cool flame propagation speeds." 44th AIAA Aerospace Sciences Meeting, Reno, NV, January 9-12, 2006. 44th AIAA Aerospace Sciences Meeting, January 2006, Paper AIAA-2006-1132. , Jan-2006
Papers from Meeting Proceedings	Foster M., Pearlman H. "The reactive-diffusive structure associated with a global heptane mechanism." Eastern States Section of the Combustion Institute, Orlando, FL, November 13-15, 2005. Eastern States Section of the Combustion Institute, November, 2005. , Nov-2005
Papers from Meeting Proceedings	Ma Y, Pearlman H. "Target and spiral waves in premixed gas combustion." The 31st International Symposium on Combustion, University of Heidelberg, Germany, August 6-11, 2006. The 31st International Symposium on Combustion, University of Heidelberg, August 2006. , Aug-2006
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Papers from Meeting Proceedings	Pearlman H, Foster M. "The cool flames experiment: Recent results at reduced and partial gravity." 42nd AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 2004. 42nd AIAA Aerospace Sciences Meeting and Exhibit, January 2004, p. 11948-11951, Paper AIAA-2004-0956. , Jan-2004